Isoliner · Topography: from DEM to catchments

A cheat sheet for the ”2. Topography” group · QGIS 3.16+ · Isoliner plugin v4 · pure NumPy, no GRASS or SAGA

Pipeline: from a DEM to Topo2Raster

Step 1. DEM - 2.01 Download DEM by extent. Set the extent, keep the defaults. Output: a 30 m terrain raster in a metric CRS.
Step 2. Contours - 1.04 Isolines from raster. Input: the DEM of step 1. A 10 m contour step, the **ELEV** field, do not raise the minimum length (short closed lines are hilltops), turn contour polygons off.
Step 3. Streamlines - 2.06 River network. Input: the DEM of step 1. Threshold 1000 cells, a denser network - a lower threshold. Line vertices already run downstream, as Topo2Raster requires.
Step 4. Summit marks - 2.09 Peaks. Input: the DEM of step 1. Radius 500 m, drop 20 m. Output: points with the **z** field - the cure for tops above the last closed contour.
Step 5. Water bodies and cliffs - 2.02 Download base topography by extent. The same extent. From OSM we take what a raster cannot give: water bodies (flat planes) and cliffs (barriers). OSM watercourses and peaks are a fallback for steps 3-4 when ele is filled.
Step 6. Terrain - 2.03 Topo2Raster. The contours of step 2 (the ELEV field), streamlines - the network of step 3, elevation points - the peaks of step 4 (the z field), lakes - the water bodies of step 5, cliffs as barriers (the step is not smeared). Cell 30 m, the extent empty. Output: the topographic terrain raster.
Control: 1.04 over the result with the same step, overlaid on the contours of step 2 - the lines must coincide. Contours from a digitized plan replace steps 1-2. All outputs land in the **Topography** group of the layer tree.

Tool reference: input, output, key parameters

Tool	Input	Output	Key parameters
2.01 Download DEM	map extent	GeoTIFF float32, meters	source: GLO-30 (DSM) or GEDTM30 (DTM, forest removed). Cell 30 m. CRS: empty = project/UTM
2.02 OSM base map	map extent	watercourses, water bodies, peaks (ele), cliffs	coastline off. Large extent: shrink it or raise the limit
1.04 Isolines from raster	DEM	lines with ELEV	step 5-10 m. Do not raise min length: short closed lines are hilltops. Moderate smoothing. Polygons off
2.03 Topo2Raster	contours (elevation field), points, streamlines, cliffs, lakes	GeoTIFF float32	points or contours required. Streamlines run downstream: OSM and 2.06 fit as is. Edge: node Z (river slope) > field elevation (plane) > shore minimum
2.04 Terrain preparation	DEM	prepared DEM	FPDEMS smoothing (edge-preserving) and/or depression filling (epsilon 0.001 for D8). Both by checkboxes
2.05 Flow D8	DEM	directions (ArcGIS codes, byte) + accumulation (cells, float32)	codes: E=1, SE=2, S=4, SW=8, W=16, NW=32, N=64, NE=128, sink=0
2.06 River network	DEM	lines: order (Strahler), acc_out, length_m	threshold = head catchment / cell area. 1000 at 30 m ≈ 0.9 km^2
2.07 Basins	DEM, pour points (opt.)	polygons: basin, area_m2 (+ label raster)	without points: from mouths by threshold. Point snap to the accumulation max, 150 m
2.08 Slope and aspect	DEM	two rasters, degrees	Horn 3×3 as gdaldem. Aspect: downslope azimuth, flats = -1
2.09 Peaks	DEM	points: z, drop	a 500 m radius suppresses secondary tops, a 20 m drop cuts bumps

Common pitfalls

- **A degree CRS.** The group works in meters. 2.01 reprojects itself: an empty target CRS = the project CRS if metric, otherwise UTM at the center. User CRSs without an EPSG code are fully supported.
- **The river threshold is tuned.** Start at 1000 and adjust threefold. A DEM check: overlay the 2.06 network on the OSM watercourses.
- **Overpass is not elastic.** Large requests get cut by OSM servers, on a failure the tool switches to a mirror. Keep the extent modest.
- **Closed basins.** Karst and subsidence troughs are erased by the hydro correction: uncheck the filling.